

Chen Zhao

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 <https://github.com/CL17189>

Education

Uppsala University, M.Sc. in Data Science-Data Engineering Sept. 2024 – Jun. 2026

- **Course Work** Database Design, Data Engineering, Statistical Machine Learning, High Performance Programming, Combinatorial Optimization, Reinforcement Learning, Advanced Probabilistic Machine Learning

Renmin University of China, Bachelor in Statistics Sept. 2020 – Jun. 2024

- **Course Work:** Probability Theory, Mathematical Statistics, Sampling Techniques, Regression Analysis, Time Series Analysis, Data Structures and Algorithms, Machine Learning, Deep Learning

Projects

IntelliQuilt - Bachelor's Thesis Github

- Developed a multimodal fusion model based on pytorch to predict sales with 90% accuracy, using multimodal data (5000+ samples) collected from a major online shopping platform.
- Conducted extensive ablation experiments to evaluate the performance of various architectures, leveraging theoretical insights from relevant literature.

Key Words: Multimodal Learning, Deep Learning, PyTorch, Ablation Study

QuantLake – Data Lakehouse Pipeline for Quantitative Analytics GitHub

- Built a scalable Delta Lake-based data lakehouse with Spark SQL and Prefect for end-to-end financial data processing.
- Developed ingestion and ETL pipelines for specific theme and fundamentals data, with partitioning, versioning, and Z-Order optimization for fast queries and time-travel analysis.
- Containerized the workflow with Docker and automated orchestration and CI/CD

Key Words: data engineering, ETL pipeline, S3, lakehouse architecture, scalable analytics

Meowmory: AI-Assisted Vocabulary Learning Application GitHub

- Developed a full-stack language learning application enabling users to memorize vocabulary with AI-generated funny stories and adaptive difficulty.
- Built a modular backend with **FastAPI**, integrating authentication, spaced-repetition scheduling, user progress tracking, and model-serving endpoints.
- Designed and deployed a **PostgreSQL**-based data schema for vocabulary, user profiles, learning history, and model outputs.
- Implemented an extensible **MCP (Model Context Protocol)** pipeline for context-aware LLM prompting and multi-step reasoning.
- Set up automated testing, versioning, and **CI/CD** workflows to ensure reliable deployment and maintainability.

Key Words: FastAPI, PostgreSQL, MCP, CI/CD, LLM integration, full-stack development

Intern Experience

DiDi, Anti-Fraud Data Analysis Intern

Mar. 2023 – Jun. 2023

- Built large-scale **data pipelines** using **Hive** and **Spark**, designing DWS-layer tables and automated **shell scripts** for daily processing (100M-level), ensuring high data integrity and timely availability.
- Conducted **behavioral data analysis** to identify fraudulent user patterns, proposed new anti-fraud **features/indicators**, and contributed to the refinement of detection models.
- Evaluated the **effectiveness and business impact** of detection strategies by designing validation metrics and deploying **monitoring dashboards** for real-time performance tracking.

Key Words: Fraud Detection, Feature Engineering, Data Pipeline, Hive, Spark

Technologies

Programming: Python, C, R, HTML/CSS

Databases & Big Data: SQL, PostgreSQL, MongoDB, Spark, Flink, Kafka

Machine Learning & Data: Scikit-learn, PyTorch, NumPy, Pandas

Tools & DevOps: Linux, Shell scripting, Git, Tableau, LaTeX